

EE582 Homework 01 Report

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Question 1

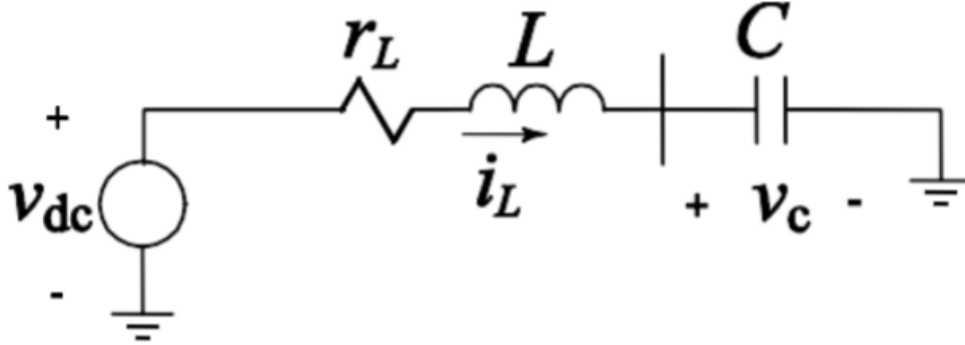


Figure 1: Circuit diagram for a simple filter for questions 1-8

A shunt harmonic filter (series RLC) is shown in Figure 1 above. It is designed for a three-phase 13.8 kV 60 Hz system to shunt the 7th harmonic current. Find the reactive power, damped and undamped resonance frequency, and quality factor of the filter. The circuit parameters are: $r_L = 1 \Omega$, $L = 19 \text{ mH}$, and $C = 8.2 \mu\text{F}$.

Step 1: Write the KVL equation:

$$V_{dc} - r_L i_L - L \frac{di_L}{dt} - v_C = 0$$

Or equivalently:

$$L \frac{di_L}{dt} + r_L i_L + v_C = V_{dc}$$

With $i_L = C \frac{dv_C}{dt}$ and $v_C = \frac{1}{C} \int i_L dt$.

Step 2: Standard second-order ODE:

$$\frac{d^2 i_L}{dt^2} + \frac{r_L}{L} \frac{di_L}{dt} + \frac{1}{LC} i_L = 0$$

Step 3: Characteristic equation:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Step 4: Calculate parameters:

$$\omega_n = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{19 \times 10^{-3} \times 8.2 \times 10^{-6}}} = 2533.5 \text{ rad/s}$$

$$\zeta = \frac{r_L}{2\sqrt{\frac{L}{C}}} = \frac{1}{2\sqrt{\frac{19 \times 10^{-3}}{8.2 \times 10^{-6}}}} = 0.0104$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 2533.3 \text{ rad/s}$$

Quality factor:

$$Q_{\text{factor}} = \frac{\omega_n L}{r_L} = \frac{2533.5 \times 0.019}{1} = 48.136$$

Eigenvalues (roots of characteristic equation):

$$\lambda = -\zeta\omega_n \pm j\omega_d = -26.316 \pm 2533.3j$$

Reactive Power Computation: The reactive power is computed for a three-phase system. First, the per-phase voltage $V_{ph} = V_{LL}/\sqrt{3}$ and current I_{ph} are calculated using the per-phase impedance $Z_{ph} = r_L + j(X_L - X_c)$. Then, the reactive power is determined as:

$$Q_{\text{kVAr}} = 3 \times |I_{ph}|^2 \times X_c$$

where $X_c = \frac{1}{2\pi fC}$ and $X_L = 2\pi fL$ at $f = 60$ Hz.

$$Q_{\text{kVAr}} = -602.04 \text{ kVAr}$$

Negative sign indicates the filter is absorbing reactive power (capacitive). All calculations confirm the system is very underdamped ($\zeta = 0.0104$).

Summary of Key Results:

- Natural frequency: $\omega_n = 2533.5 \text{ rad/s}$
- Damping ratio: $\zeta = 0.0104$
- Damped frequency: $\omega_d = 2533.3 \text{ rad/s}$
- Quality factor: $Q = 48.136$
- Eigenvalues: $-26.316 \pm 2533.3j$
- Reactive power: -602.04 kVAr

Question 2

Calculate the transient response of the circuit shown in Fig. 1(a) to a step voltage change if $v_{dc}(t) = s(t) = 11.3 \text{ kV}$ for $t \geq 0$. Comment on the eigenvalues and response type.

Referring again to the series RLC circuit in Q1 (Figure 1) and writing relevant KVL and current equations, we have:

KVL:

$$V_{dc} - r_L i_L - L \frac{di_L}{dt} - v_C = 0 \quad (1)$$

where i_L is the inductor current and v_C is the capacitor voltage.

Capacitor current:

$$i_L = C \frac{dv_C}{dt} \quad (2)$$

Substitute (2) into (1):

$$V_{dc} - r_L C \frac{dv_C}{dt} - LC \frac{d^2 v_C}{dt^2} - v_C = 0$$

Rearrange to standard form:

$$LC \frac{d^2 v_C}{dt^2} + r_L C \frac{dv_C}{dt} + v_C = V_{dc}$$

Divide both sides by LC :

$$\frac{d^2 v_C}{dt^2} + \frac{r_L}{L} \frac{dv_C}{dt} + \frac{1}{LC} v_C = \frac{V_{dc}}{LC}$$

This is a standard second-order nonhomogeneous ODE. The homogeneous solution is:

$$\frac{d^2 v_C}{dt^2} + \frac{r_L}{L} \frac{dv_C}{dt} + \frac{1}{LC} v_C = 0$$

Eigenvalue computation: The characteristic equation is:

$$s^2 + \frac{r_L}{L} s + \frac{1}{LC} = 0$$

The eigenvalues are:

$$s = \frac{-\frac{r_L}{L} \pm \sqrt{\left(\frac{r_L}{L}\right)^2 - 4 \cdot \frac{1}{LC}}}{2}$$

For the given values ($r_L = 1 \text{ } \Omega$, $L = 19 \text{ mH}$, $C = 8.2 \text{ } \mu\text{F}$):

$$\begin{aligned} \frac{r_L}{L} &= \frac{1}{0.019} = 52.63 \text{ s}^{-1} \\ \frac{1}{LC} &= \frac{1}{0.019 \times 8.2 \times 10^{-6}} = 6,430,041 \text{ s}^{-2} \end{aligned}$$

Plug into the quadratic formula:

$$s = \frac{-52.63 \pm \sqrt{(52.63)^2 - 4 \times 6,430,041}}{2}$$

The discriminant is negative, so the roots are complex conjugates:

$$s = -26.316 \pm j 2533.3$$

Interpretation: The eigenvalues are complex conjugates with negative real parts, confirming the circuit is underdamped. Based on the very low value of the damping ratio $\zeta = 0.0104$ calculated in Q1, the transient response will exhibit oscillatory behavior with a slow decay.

General solution:

$$v_C(t) = v_{C,\infty} + [v_{C,0} - v_{C,\infty}] \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \right]$$

where $v_{C,\infty}$ is the steady-state value, $v_{C,0}$ is the initial value, and $\omega_d = \omega_n \sqrt{1 - \zeta^2}$.

For a step input, $v_{C,\infty} = x_{SS}$, and the solution simplifies to:

$$x(t) = x_{SS} \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \right]$$

For the given values ($r_L = 1 \Omega$, $L = 19 \text{ mH}$, $C = 8.2 \mu\text{F}$), the numeric solution is:

$$x(t)[kV] = 11.3 * [1 - 1.00005 e^{-26.34 t} \sin(2533.3 t + 1.5604)]$$

Question 3

Develop a Simulink model of the circuit shown in Fig. 1(a) using conventional Simulink blocks (i.e., summers, gains, integrals, etc.). Compute the transient response (solutions) on time interval $[0, 0.1]$ s using the ode3 solver with 0.1 ms time-step size.

Block Diagram

The following block diagram represents the Simulink implementation of the RLC circuit for part (a):

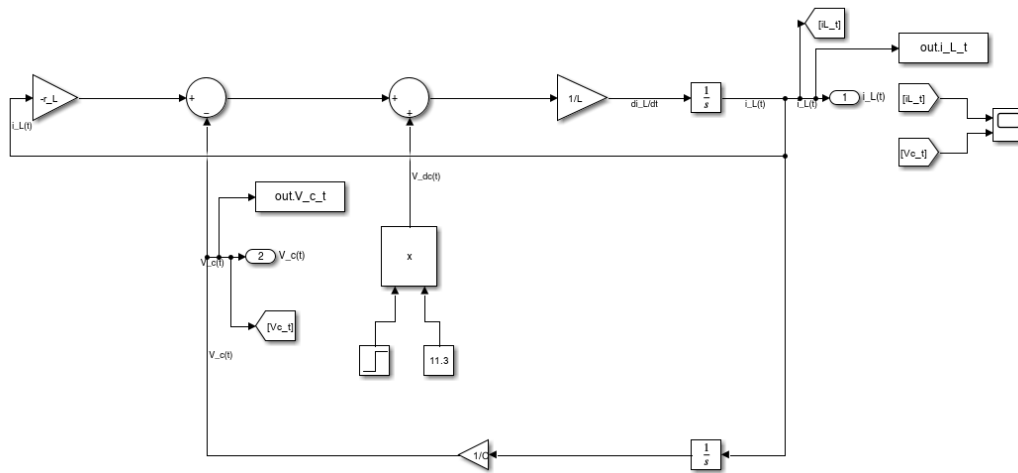


Figure 2: Simulink block diagram for the series RLC circuit (part a).

Simulation Results

The transient response of the circuit was simulated in Simulink using the `ode3` solver with a fixed time step of 0.1 ms over the interval $[0, 0.1]$ s. The following plot shows the inductor current $i_L(t)$ and capacitor voltage $v_C(t)$:

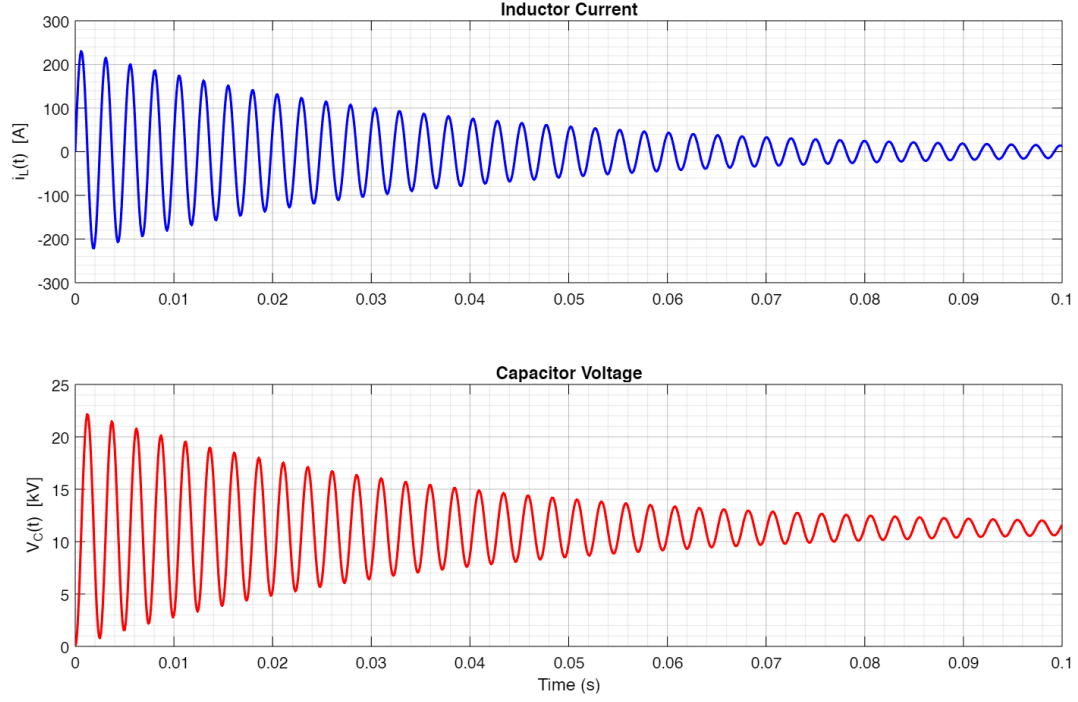


Figure 3: Simulink simulation results: $i_L(t)$ and $v_C(t)$ for the RLC circuit.

Discussion

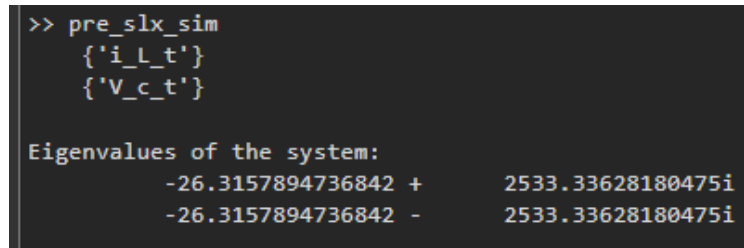
The simulation results show the expected underdamped oscillatory response, consistent with the analytical solution derived in Q1 and Q2. The current and voltage waveforms exhibit oscillations with a slow decay, as predicted by the low damping ratio ($\zeta = 0.0104$). The Simulink model accurately captures the circuit dynamics using only basic blocks and the specified solver settings.

Question 4

Find the system eigenvalues using the MATLAB functions `linmod` and `eig` on the command line. Compare the results with part 2.

MATLAB Output

The following figure shows the eigenvalues computed using MATLAB's `linmod` and `eig` functions:



```
>> pre_slx_sim
    {'i_L_t'}
    {'V_c_t'}

Eigenvalues of the system:
      -26.3157894736842 +      2533.33628180475i
      -26.3157894736842 -      2533.33628180475i
```

Figure 4: Eigenvalues of the system computed using `linmod` and `eig` in MATLAB.

Comparison with Analytical Results

The eigenvalues obtained from MATLAB are:

$$-26.3158 \pm 2533.34i$$

These match very closely with the hand-computed values in Q2:

$$-26.316 \pm 2533.3i$$

The small numerical difference is due to rounding and floating-point precision in MATLAB. Both methods confirm the system is underdamped with complex conjugate eigenvalues, as expected from the analytical derivation.

Question 5

Analyze the stability of the Forward Euler (ode1) and explicit RK5 (ode5) methods for the RLC circuit. Plot the stability regions and indicate the range of h for which the methods are stable.

Stability Plots

The following figure shows the stability regions for Forward Euler (ode1) and explicit RK5 (ode5) methods, with the scaled eigenvalues $z = h\lambda$ plotted for various step sizes h :

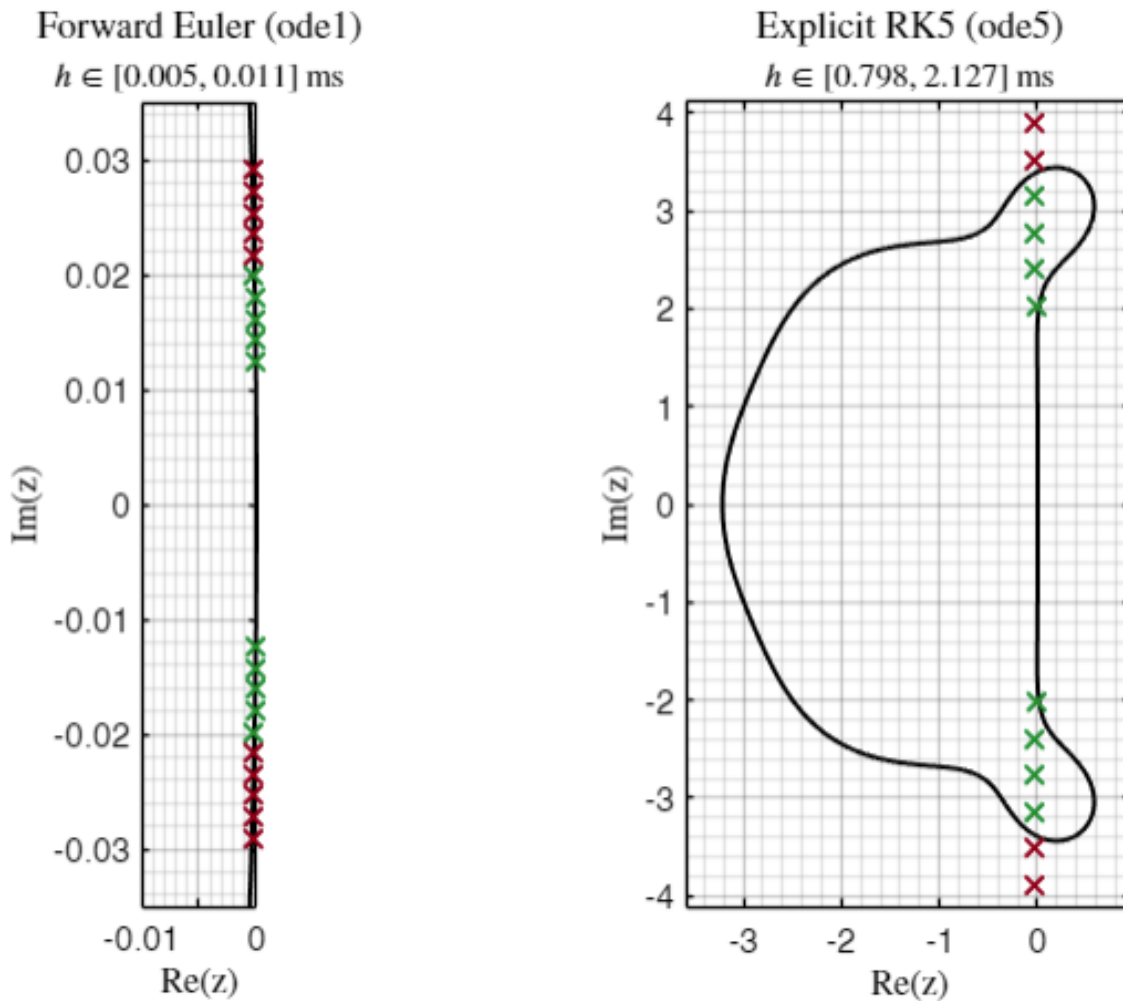


Figure 5: Stability regions for Forward Euler (ode1, left) and explicit RK5 (ode5, right). Each cross represents $z = h\lambda$ for a candidate step size h and the system eigenvalues λ . Green crosses are inside the stability region (stable), red crosses are outside (unstable). Note that the plot for Forward Euler is zoomed in to show the small stability region.

Stability Criteria and z Formulae

For a linear system $\dot{x} = \lambda x$, the stability of a numerical method is determined by the location of $z = h\lambda$ in the method's stability region:

- **Forward Euler (FE):** The stability function is $R(z) = 1 + z$. The method is stable if $|1 + z| < 1$.
- **Explicit RK5 (Taylor-5):** The stability function is $R(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \frac{z^5}{5!}$. The method is stable if $|R(z)| < 1$. (Note: MATLAB's ode5 is not exactly a Taylor-5 method, but the Taylor-5 region is a good approximation for illustration.)

Interpretation of Crosses

Each cross in the plot corresponds to a value of $z = h\lambda$ for a particular step size h and one of the system's eigenvalues λ .

Green crosses indicate z values inside the stability region (method is stable for that h), while **red crosses** are outside (unstable for that h).

Maximum Stable Step Size

From the plot and MATLAB output:

- **Euler:** $h_{\max} \approx 0.008$ ms
- **Taylor-5:** $h_{\max} \approx 1.329$ ms

These values indicate the largest time step for which each method remains stable for the given system.

Summary

The explicit RK5 (Taylor-5) method allows for a much larger stable time step than Forward Euler, as seen from the much larger stability region. The actual ode5 implementation in MATLAB is not exactly Taylor-5, but the qualitative stability behavior is similar for this analysis.

Question 6

Use any one of the fixed-time-step solvers of your own choice. Plot the solution for i_L and v_C . Experimentally establish the largest time step $h_{\max}$ that produces a stable and convergent solution that you can trust. Compare your result with the result of using the ode5 at $h = 0.5 \times 10^{-3}$ s and $h = 1.0 \times 10^{-3}$ s.

Simulation Results

The following figures show the results of the fixed-step simulations:

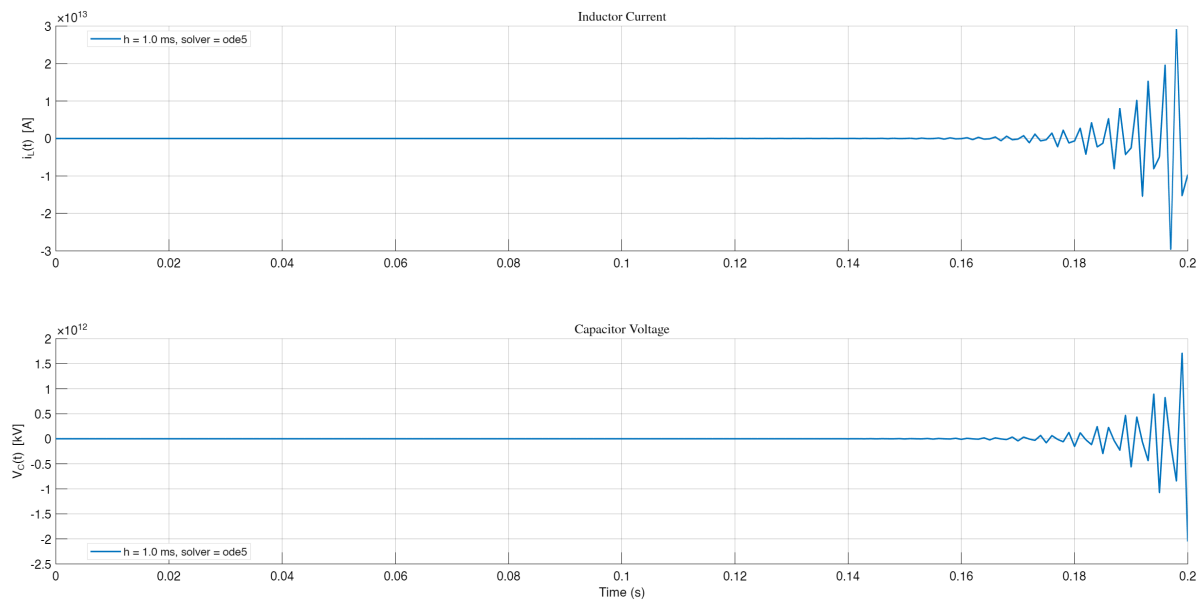


Figure 6: Diverging solution for $h = 1$ ms using ode5 (RK5).

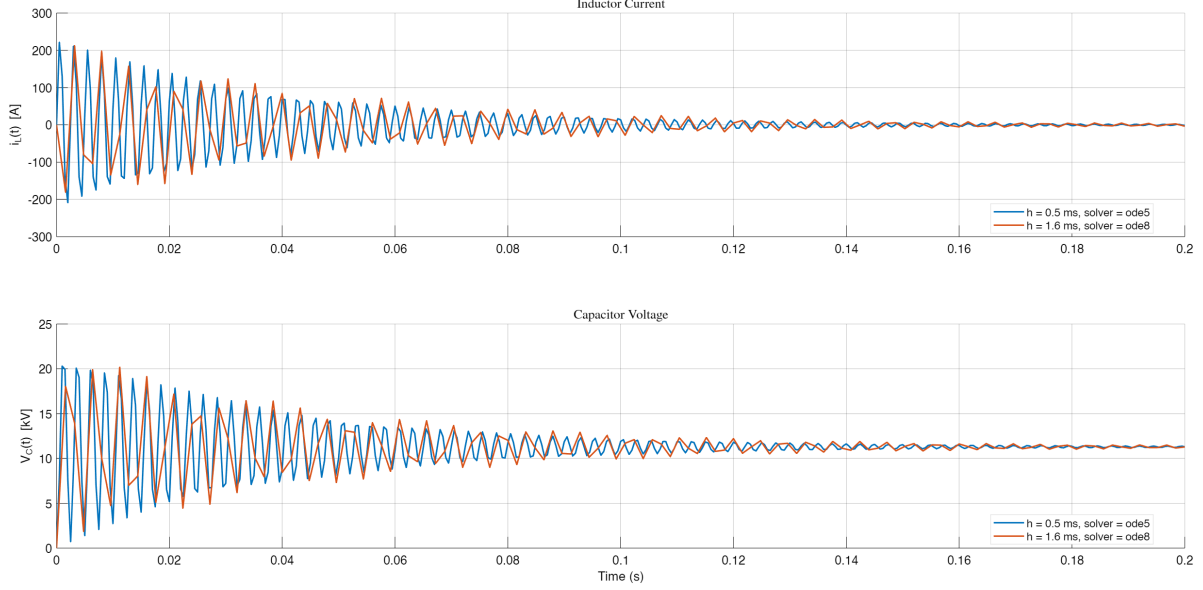


Figure 7: Converging solutions for RK8 with $h = 1.6$ ms and ode5 with $h = 0.5$ ms. The legend indicates the solver and step size.

Discussion

For this part, I used an explicit RK8 method for my own solver. The maximum time step $h_{\max}$ for which the RK8 method produced a stable and accurate solution was determined experimentally to be 1.6 ms. For ode5 (RK5), the $h = 1$ ms case is shown separately above, as it diverges and would spoil the scale of the converging plot.

The experimentally obtained $h_{\max}$ for RK8 was found to be less than the theoretical value for Taylor-5 (≈ 1.3 ms). For ode5 (RK5), the maximum stable h was found to be less than 0.7 ms (figure not shown here), which is also less than the theoretical prediction. The theoretical values are based on the Taylor-5 stability region, but ode5 is not exactly Taylor-5, and RK8 has a different (larger) stability region.

Remark

The fact that the experimentally determined $h_{\max}$ is smaller than the theoretical value is mainly due to (i) the difference between the actual Dormand–Prince implementation of ode5 and the idealized Taylor-5 surrogate (their stability regions are not identical), and (ii) the oscillatory eigenvalues of the system, where reliable simulation typically requires ~ 20 – 30 steps per cycle. For our case, $f_d \approx 403$ Hz so $T = 1/f_d \approx 2.48$ ms; thus an accuracy-driven step is

$$h \lesssim \frac{T}{20-30} \approx 0.08\text{--}0.12 \text{ ms}.$$

By contrast, a theoretical $h_{\max}$ inferred from the Taylor-5 absolute-stability boundary is much larger (on the order of ~ 0.5 – 1.0 ms in our plots), which explains the discrepancy with the empirically safe step used in simulation.

Question 7

Using a variable-time-step solver of your choice, establish a convergent solution that you can trust. Plot the solution i_L and v_C , and the time step h as functions of time. Your results should be accurate enough to be assumed as reference solution.

Discussion

Since I initially don't have a notion of what a 'good' solver output looks like for this problem, the only 'reference' transient response I'm assuming is the one I can obtain by plotting the transient function derived in Q2 – I use that as a base for comparison in Q8.

After experimenting with all variable step solvers in MATLAB (ode45, ode23, ode113, ode15s, ode23s, ode23t, ode23tb), my findings are somewhat inconclusive. I personally liked stiff variations of ode23 (ode23s, ode23t, ode23tb). All subsequent comparison and discussion, including the reasoning behind my choice is in Q8.

Question 8

Using variable-step solvers, set $h_{max} = 1 \times 10^{-2}$ and both error tolerances $\varepsilon_{abs} = \varepsilon_{rel} = 10^{-3}$. Experiment with all the available variable-step solvers. In a table, print the maximum time-step size h_{max} chosen by each solver and the total number of steps taken. Comment on how these solutions compare with the reference solution found in part 7. Which solver in your opinion is more efficient and does the best job for this system? Compare the step-size and the overall number of steps observed in parts 6, 7, and this part.

Figures and Analysis

The following figures illustrate the transient response of the inductor current $i_L(t)$ and capacitor voltage $V_C(t)$ for both nonstiff and stiff solvers. Figure 8 includes all solvers, highlighting the performance of solvers that fail to dampen oscillations effectively. Figure 9 excludes these solvers, focusing only on the solvers that converge to the expected damped sinusoidal response:

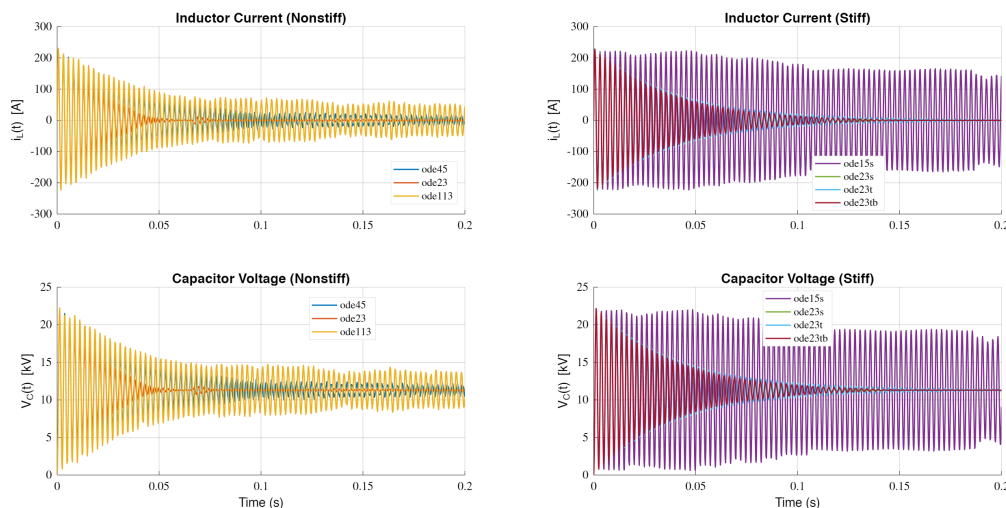


Figure 8: Transient Response of Inductor Current and Capacitor Voltage (Including Non-Dampening Solvers)

Table of Results

The table below, supported by fig. 10, summarizes the performance of each solver, including the maximum step size used, the number of steps taken, the simulation time, and the RMSE error:

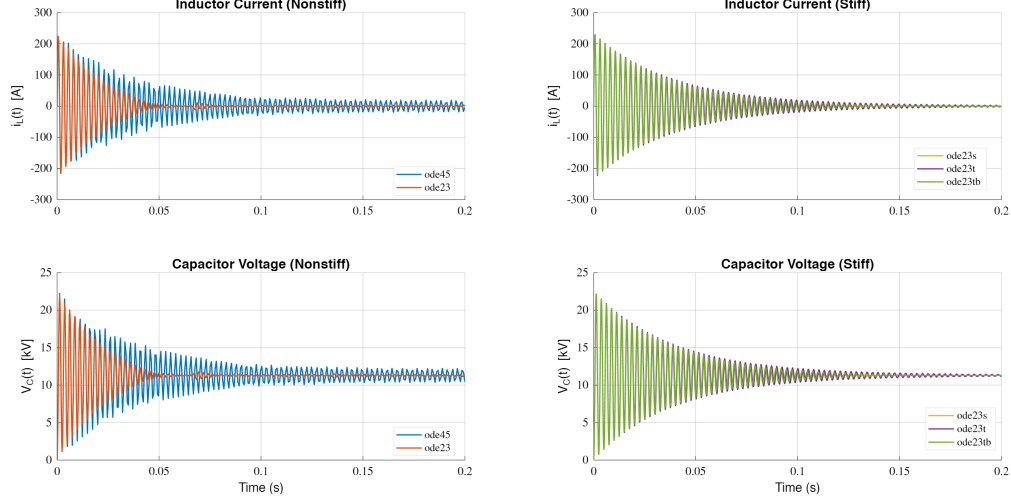


Figure 9: Transient Response of Inductor Current and Capacitor Voltage (Excluding Non-Dampening Solvers)

Solver	Type	MaxStepUsed	NumSteps	SimTime_sec	Error_kV
{'ode45' }	"Nonstiff"	0.00085483	318	7.7641	0.61709
{'ode23' }	"Nonstiff"	0.0013118	415	0.4192	1.2333
{'ode113' }	"Nonstiff"	0.00061603	755	0.37917	1.5953
{'ode15s' }	"Stiff"	0.00031737	1174	0.2778	5.3695
{'ode23s' }	"Stiff"	0.01	468	0.49326	1.9883
{'ode23t' }	"Stiff"	0.00072387	784	0.23214	2.1127
{'ode23tb' }	"Stiff"	0.01	535	0.2111	1.7507

Figure 10: Simulation Results for Variable-Step and Fixed-Step ODE Solvers

Solver	Type	h_{Max}	Num Steps	Time [s]	Error [kV]
ode45	Nonstiff	0.0008548	318	0.2753	0.6171
ode23	Nonstiff	0.0013118	415	0.3346	1.2333
ode113	Nonstiff	0.0006160	755	0.2697	1.5953
ode15s	Stiff	0.0003174	1174	0.3068	5.3695
ode23s	Stiff	0.01	468	0.3072	1.9883
ode23t	Stiff	0.0007239	784	0.2488	2.1127
ode23tb	Stiff	0.01	535	0.2490	1.7507
ode5	Fixed-step	0.0005	401	0.2556	0.2283
ode8	Fixed-step	0.0005	401	0.2388	0.0000422

Table 1: Summary of Solver Performance Including Fixed-Step RK8

Discussion

Nonstiff vs Stiff Solvers

While nonstiff solvers (ode45, ode23) generally showed lesser error, their transient response 'shape' was less consistent with a damped sinusoid as expected from the analytical solu-

tion. Stiff solvers (ode15s, ode23s) produced responses that visually aligned better with the expected damped sinusoidal shape, but they exhibited higher numerical error.

Maximum step size not necessarily proportional to error

Interestingly, nonstiff solver ode113 and stiff solver ode15s took very small steps yet resulted in a significantly higher error compared to other solvers. On the other hand, stiff solvers like ode23s and ode23tb took the largest possible steps (perhaps only occasionally or regularly), yet managed to achieve lower errors. This behavior warrants a deeper analysis of how these solvers handle step size and error accumulation.

Fixed-Step Solvers

Fixed-step solvers RK5 and RK8 demonstrated exceptional performance, with RK8 achieving the lowest error (0.0000422 kV). However, the fixed step size of 0.5 ms is overly generous compared to the variable-step solvers, making the comparison somewhat unfair. A more balanced comparison would involve reducing the fixed step size to align with the typical step sizes observed in variable-step solvers. As previously observed, a step size of ≥ 0.7 ms is infeasible for RK5, so a variable step size solver taking a maximum step size of 10 ms and still not diverging is in a way already superior.

Simulation Time Analysis

Moreover, the simulation times across all solvers were found to vary significantly between runs, making solve time an inconclusive metric for this system. The choice of solver should therefore depend more on the error metric and the qualitative shape of the transient response rather than computational efficiency. Among the variable step size solvers, stiff solvers like ode23s, ode23t and ode23tb remain reliable for capturing the expected response shape.

Remarks on Error Computation

The error metric used in this analysis is the root mean square error (RMSE), computed by evaluating the analytical solution at the same time steps as those of the numerical solver. The analytical solution, derived in Q2, is given by:

$$V_c(t)[kV] = 11.3 [1 - 1.00005 e^{-26.34t} \sin(2533.3t + 1.5604)]$$

This approach ensures a fair comparison, as the error is calculated point-by-point at identical time instances.

Conclusion - Recommended Solver

In conclusion, the stiff solvers, particularly the ode23 variants, are recommended for this problem due to their ability to handle rapid dynamics efficiently while maintaining accuracy.

Question 9

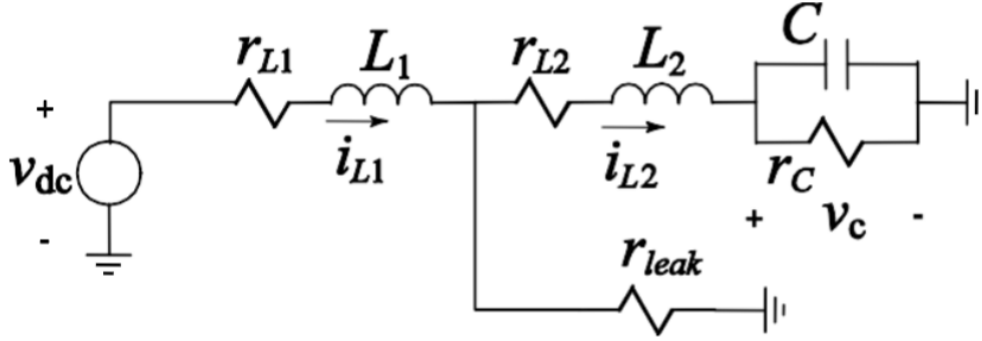


Figure 11: Circuit diagram of the full filter for questions 9-12

In Fig. 1(a) the equivalent series resistance and inductance of the capacitor (r_{L2} and L_2) were added to the inductance parameters r_{L1} and L_1 (i.e. $r_L = r_{L1} + r_{L2}$ and $L = L_1 + L_2$). In Fig. 1(b), they are separated and also the capacitor leakage resistance (r_c) is considered. In addition, the discharge resistance (r_{leak}) is shown in Fig. 1(b). This resistance is used to discharge the filter capacitor when it is off-line for safety purposes which was ignored in the simple model shown in Fig. 1(a). Implement the model of Fig. 1(b) in Simulink using basic blocks considering $r_{L2} = 323 \text{ m}\Omega$, $L_2 = 1 \text{ mH}$, $r_c = 1 \text{ M}\Omega$, and $r_{leak} = 30 \text{ k}\Omega$. If the filter is disconnected, how long does it take for the capacitor to fully discharge?

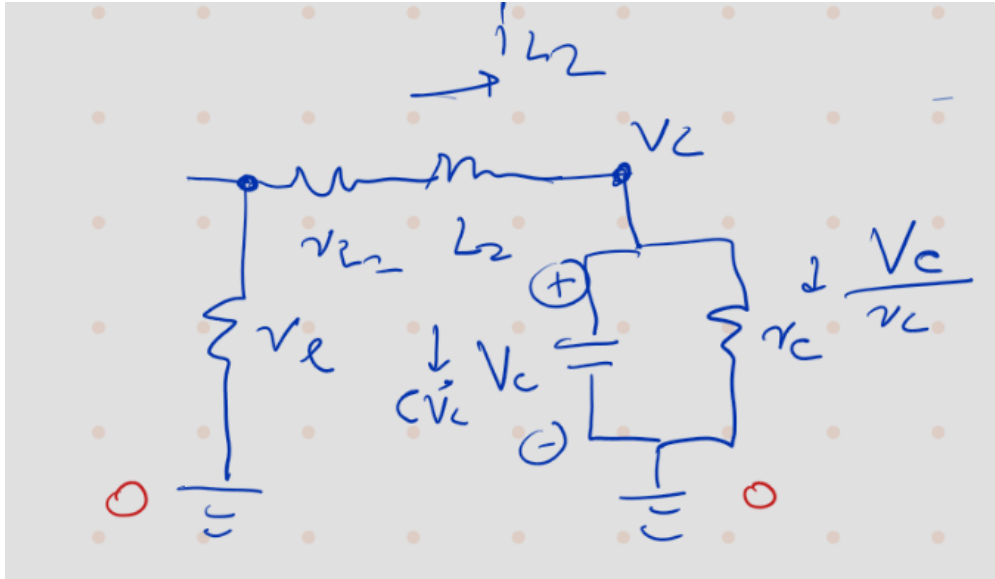


Figure 12: Hand-drawn Circuit Diagram for Disconnected Filter Analysis

The derivation below refers to Figure 12 for analyzing the disconnected filter. The capacitor C discharges through its leakage resistance r_c and the discharge resistor r_{leak} . The relevant equations can be derived as follows.

KCL at the capacitor node:

$$i_{L2} = C \frac{dv_C}{dt} + \frac{v_C}{r_C} \quad (1)$$

KVL across L_2 and resistances:

$$-(r_{\text{leak}} + r_{L2})i_{L2} - L_2 \frac{di_{L2}}{dt} = v_C \quad (2)$$

Substitute (1) into (2):

$$-(r_{\text{leak}} + r_{L2}) \left(C \frac{dv_C}{dt} + \frac{v_C}{r_C} \right) - L_2 \left(C \frac{d^2v_C}{dt^2} + \frac{1}{r_C} \frac{dv_C}{dt} \right) = v_C$$

Rearrange terms:

$$-L_2 C \frac{d^2v_C}{dt^2} - \left[(r_{\text{leak}} + r_{L2})C + \frac{L_2}{r_C} \right] \frac{dv_C}{dt} - \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1 \right) v_C = 0$$

Multiply through by -1 :

$$L_2 C \frac{d^2v_C}{dt^2} + \left[(r_{\text{leak}} + r_{L2})C + \frac{L_2}{r_C} \right] \frac{dv_C}{dt} + \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1 \right) v_C = 0$$

Divide through by $L_2 C$:

$$\frac{d^2v_C}{dt^2} + \left(\frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C} \right) \frac{dv_C}{dt} + \frac{1}{L_2 C} \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1 \right) v_C = 0$$

Final second-order ODE (disconnected case):

$$\boxed{\frac{d^2v_C}{dt^2} + \left(\frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C} \right) \frac{dv_C}{dt} + \frac{1}{L_2 C} \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1 \right) v_C = 0}$$

This is a homogeneous second-order differential equation governing the capacitor voltage $v_C(t)$ during discharge when the filter is disconnected. The eigenvalues can be obtained by solving the characteristic quadratic:

$$s^2 + \alpha s + \beta = 0,$$

where

$$\alpha = \frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C}, \quad \beta = \frac{1}{L_2 C} \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1 \right).$$

Disconnected Filter Analysis - Overdamped Case

The disconnected filter is governed by the second-order differential equation derived earlier. Solving the characteristic equation:

$$s^2 + \alpha s + \beta = 0,$$

where

$$\alpha = \frac{r_{\text{leak}} + r_{L2}}{L_2} + \frac{1}{r_C C},$$

$$\beta = \frac{1}{L_2 C} \left(\frac{r_{\text{leak}} + r_{L2}}{r_C} + 1 \right),$$

we obtain the poles:

$$s_1 = -30000318.9 \text{ s}^{-1}, \quad s_2 = -4.18694867 \text{ s}^{-1}.$$

The time constants associated with these poles are:

$$\tau_{\text{fast}} = \frac{1}{|s_1|} = 3.333 \times 10^{-8} \text{ s (0.000 ms)},$$

$$\tau_{\text{slow}} = \frac{1}{|s_2|} = 0.238837 \text{ s (238.837 ms)}.$$

Since the system is stiff, the fast-decaying pole s_1 can be ignored for practical purposes. The discharge time is determined by the slower time constant τ_{slow} . For approximately 99.3% discharge, the total time is:

$$\tau_{\text{discharge}} = 5 \times \tau_{\text{slow}} = \boxed{1.194 \text{ s (1194 ms)}}.$$

Remark: The large damping ratio ($\zeta = 1338.39$) confirms the overdamped nature of the system, making the slower pole dominant for discharge calculations.

Question 10

Question 11

Question 12